TASK 1 – ML CLASSIFICATION PROJECT: STUDENT PERFORMANCE PREDICTION

OBJECTIVE:

***To predict whether a student will pass or fail using machine learning techniques.**

TOOLS USED:

***Python**

***Pandas**

***Matplotlib**

***Scikit-learn**

***Google Colab**

DATASET DETAILS:

***Data name: Student performance dataset**

***Number of rows:395**

***Number of columns:33**

***Missing values:0**

```
Archive:  student+performance.zip
replace .student.zip_old? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
 extracting: .student.zip_old
 extracting: student.zip
Archive:  student.zip
replace student-mat.csv? [y]es, [n]o, [A]ll, [N]one, [r]ename:   school sex  age address famsize Pstatus  Medu  Fedu     Mjob      Fjob  ... \
0     GP   F   18       U     GT3       A     4     4  at_home   teacher  ...
1     GP   F   17       U     GT3       T     1     1  at_home     other  ...
2     GP   F   15       U     LE3       T     1     1  at_home     other  ...
3     GP   F   15       U     GT3       T     4     2   health  services  ...
4     GP   F   16       U     GT3       T     3     3    other     other  ...

  famrel freetime  goout  Dalc  Walc health absences  G1  G2  G3
0      4        3      4     1     1      3        6   5   6   6
1      5        3      3     1     1      3        4   5   5   6
2      4        3      2     2     3      3       10   7   8  10
3      3        2      2     1     1      5        2  15  14  15
4      4        3      2     1     2      5        4   6  10  10

[5 rows x 33 columns]
```

```
Dataset Shape: (395, 33)

Column Names:
Index(['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu',
       'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime',
       'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery',
       'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc',
       'Walc', 'health', 'absences', 'G1', 'G2', 'G3'],
      dtype='object')

Missing Values:
school        0
sex           0
age           0
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
guardian      0
traveltime    0
studytime     0
failures      0
schoolsup     0
famsup        0
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64
```

DATA VISUALIZATION:

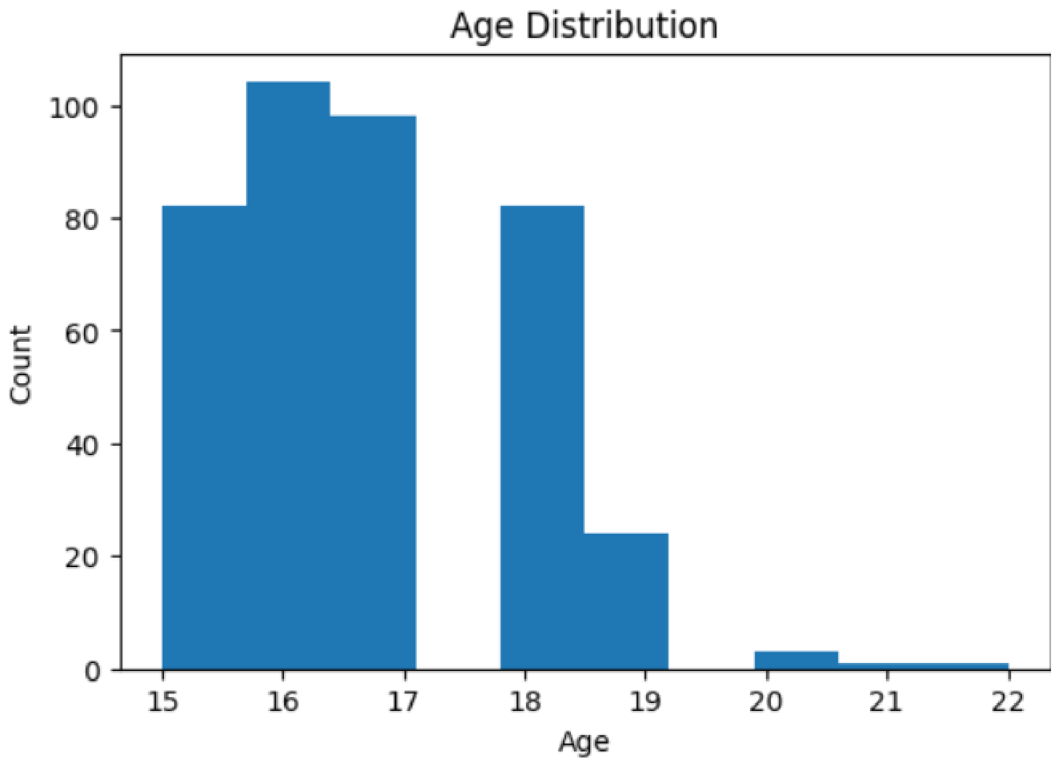
***The following visualization were created:**

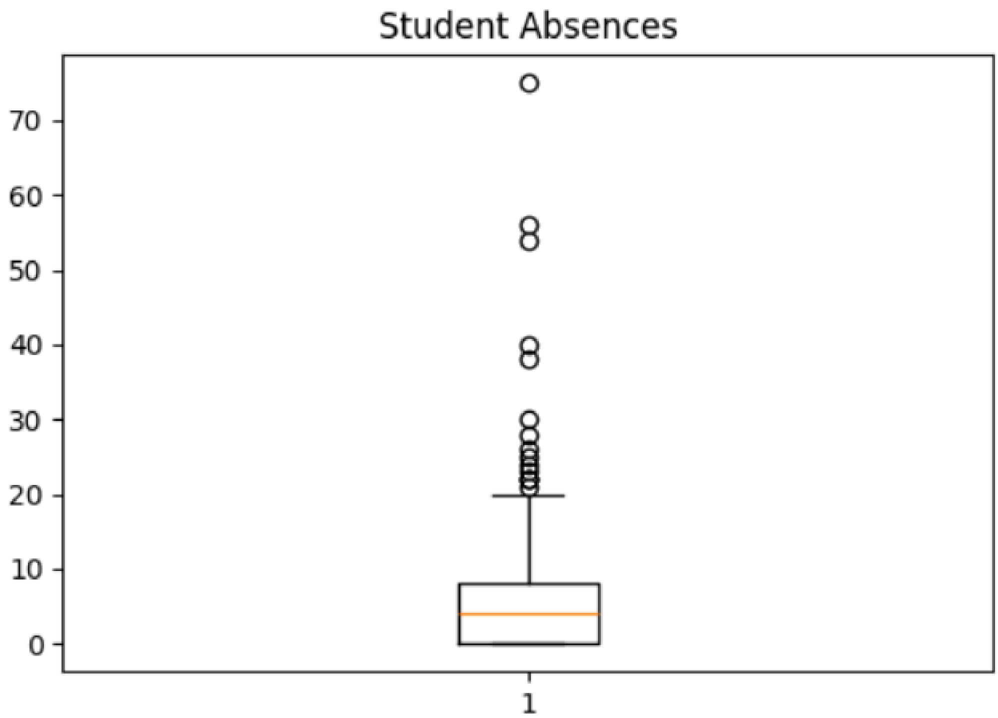
***Histogram**

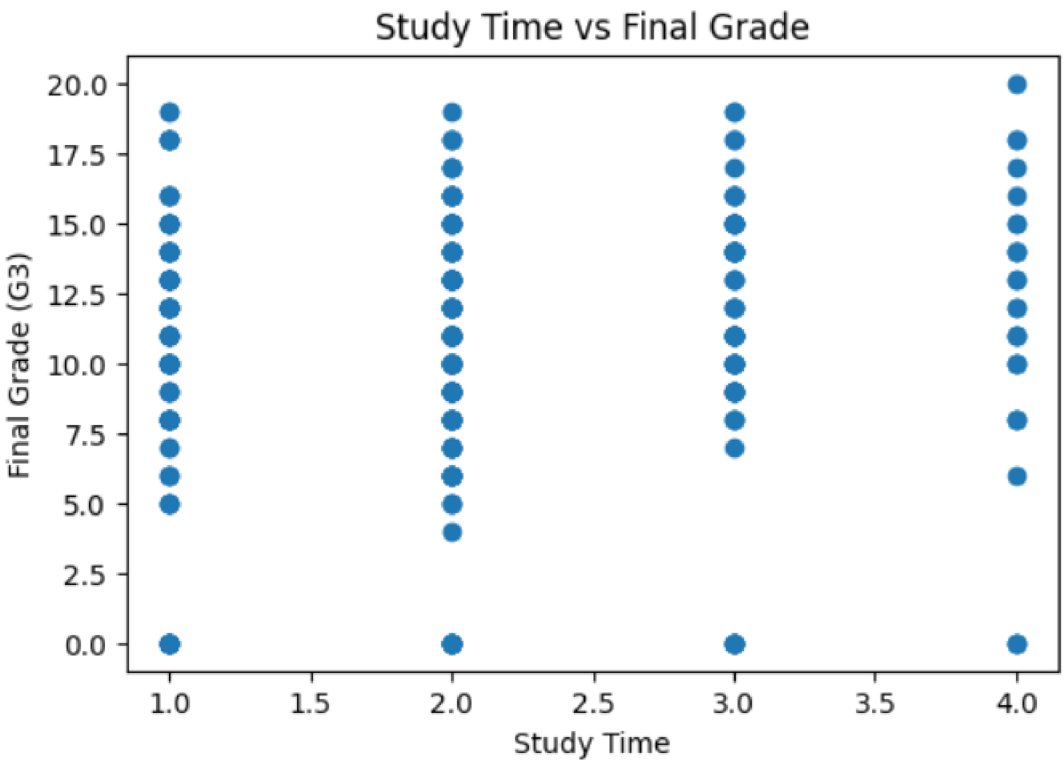
***Box plot**

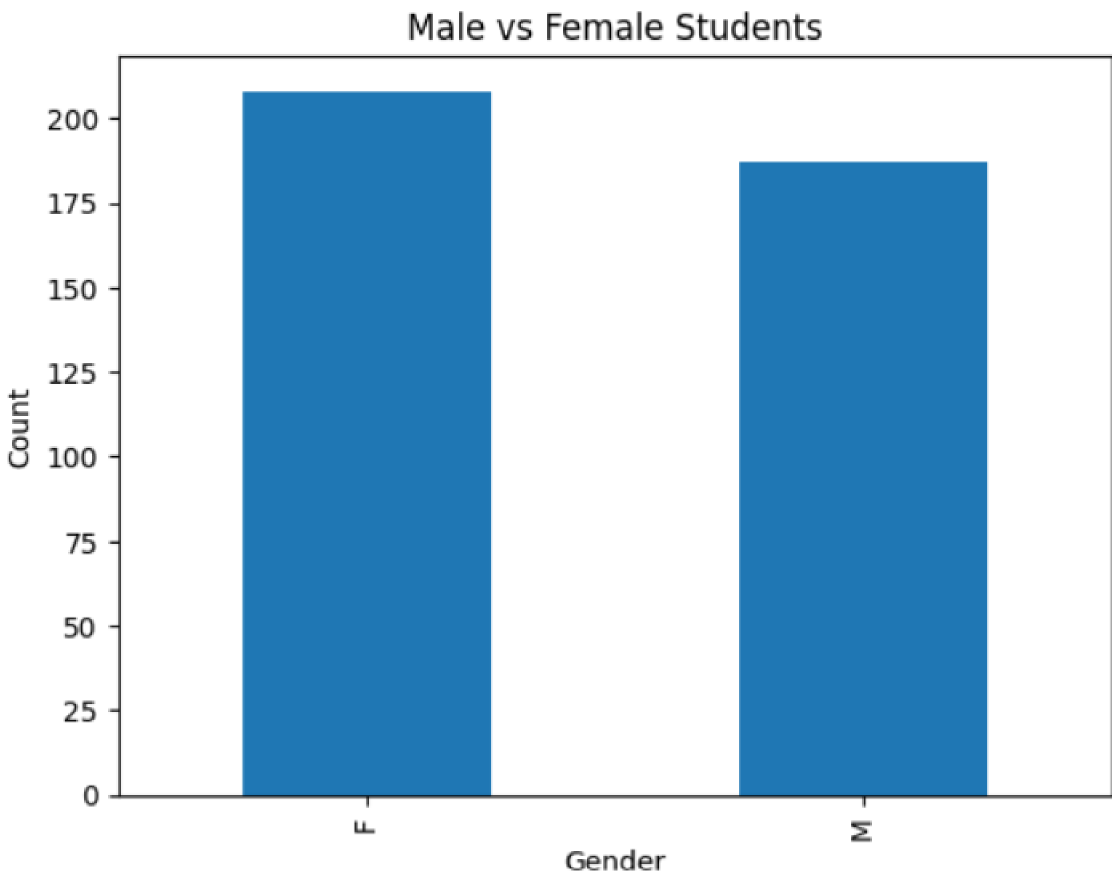
***Scatter plot**

***Bar chart**


Age Distribution
Count
100
80
60
40
20
0
15
16
17
18
19
20
21
22
Age


Student Absences
70
60
50
40
30
20
10
0
1





MODEL USED:

***Logistic Regression**

RESULT:

```
Accuracy: 73.42 %
```

CONCLUSION:

***The Machine Learning model successfully predicted whether a student would pass or fail with an accuracy of 73.42%.**